

PREDICTING DEPARTMENT-LEVEL PATIENT LOAD IMBALANCE IN HOSPITALS

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Problem Statement

Hospitals often face overcrowding in specific departments such as Emergency, ICU, or Wards due to uneven patient flow and Delayed discharges.



Motivation

- Hospitals frequently face department overcrowding, which negatively impacts patient care and staff efficiency.
- Most healthcare AI solutions focus on disease prediction, while hospital operations and patient flow challenges remain underexplored.
- This project enabled me to build an end-to-end data and AI solution that addresses a real-world operational problem.

Objectives

- To design a hospital-agnostic decision support system for predicting department-level patient load imbalance.
- To use public, anonymized datasets that reflect real hospital operational behavior.
- To identify generalizable patient flow patterns, not hospital-specific details.
- To build a scalable and privacy-aware pipeline that can be adapted to real hospital data in the future.

Scope

- Analysis of emergency, admission, and discharge data
- Prediction of future department overload risk
- Visualization of insights using dashboards and SQL queries
- Support for operational decision- making not clinical diagnosis

Literature Survey — Real-World Examples

Situation / Example	What Happened	Why It Matters (Relation to Project)
Connecticut ER boarding delays	~39% of admitted patients waited ≥4 hours in ER due to lack of beds; boarding is common and increasing.	Shows real operational overload; underscores need for <i>predictive planning</i> rather than reactive monitoring.
Baptist Health Medical Center ER overcrowding	Rising ER boarding (post-pandemic) strained capacity, delayed care, diverted ambulances, and increased patient risk.	Confirms that <i>emergency department crowding is an ongoing operational challenge</i> , not just a historical issue.
ED overcrowding research evidence	Studies show emergency department crowding causes long waits, higher mortality/morbidity, and system inefficiency.	Academic confirmation that <i>overcrowding is a persistent, systemic problem</i> that data + predictive methods should address.
Blacktown Hospital (Australia)	Patients waited >24 hrs on emergency floors or in corridors due to capacity shortages; public concern and political attention.	Real hospital overload case; shows <i>department load impacts quality of care and public trust</i> .

General Research

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How one hospital combated ER overcrowding with predictive analytics

Data-informed throughput optimization created the equivalent capacity of a 30-bed unit, according to the hospital president.

By Elizabeth Stricker, BSN, RNPublished: 30 Oct 2025



When hospital beds aren't available, admitted patients stay in the emergency department. Care is delayed and ambulances get diverted.

Admission delays harm patients, too. According to a [systematic review](#) published in *Critical Care Medicine*, prolonged ED boarding leads to longer ICU stays, more mechanical ventilation, worse organ dysfunction and significantly higher mortality. Another [study showed](#) stroke patients who were boarded for more than 24 hours had a 57% mortality rate, compared to 38% for those admitted to the ICU immediately.

ED boarding [is on the rise](#) — a crisis Baptist Health Medical Center in North Little Rock experienced firsthand.



10 use cases for predictive analytics in healthcare
Read Here



Health Tech's Role in Addressing Social Determinants of Health
-PointClickCare



How consulting supports improved revenue cycle management

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Blacktown Hospital

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Coordinates:  33.7756°S 150.9175°E

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Services

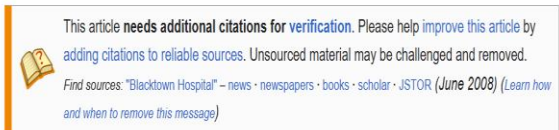
History

- Clinical Research
- Expansion Project

Controversies and notable incidents

- Emergency department overcrowding and wait times (2022–2025)
- Patient experience and performance ratings
- Maternity services and newborn deaths (2019–2020)
- Discharge decisions and family complaints
- Clinical safety reviews

References



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Find sources: "Blacktown Hospital" – news · newspapers · books · scholar · JSTOR (June 2008) ([Learn how and when to remove this message](#))

Blacktown Hospital is a university teaching hospital in Blacktown, New South Wales, Australia, about 34 kilometres from the Sydney CBD. Together with Mount Druitt Hospital and associated community health centres,^[1] it is a part of the Western Sydney Local Health District (WSLHD).^[2] The hospital is located in one of the fastest population growth areas in NSW, caring for patients from highly diverse cultural and social-economic status backgrounds.

Services [edit]

Blacktown Hospital provides a wide range of health services including:^[3]

- Acute Medical

Blacktown Hospital

Western Sydney Local Health District



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General Research

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News > Health News

HEALTH ALERT 'Extremely busy' warning issued amid overcrowding crisis as HSE issue 'consider all care options' reminder over hospital

Sixty-one patients are currently on trolleys waiting for a bed at CUH

Isabela Boechat, News Reporter
Published: 16:19, 20 Oct 2025



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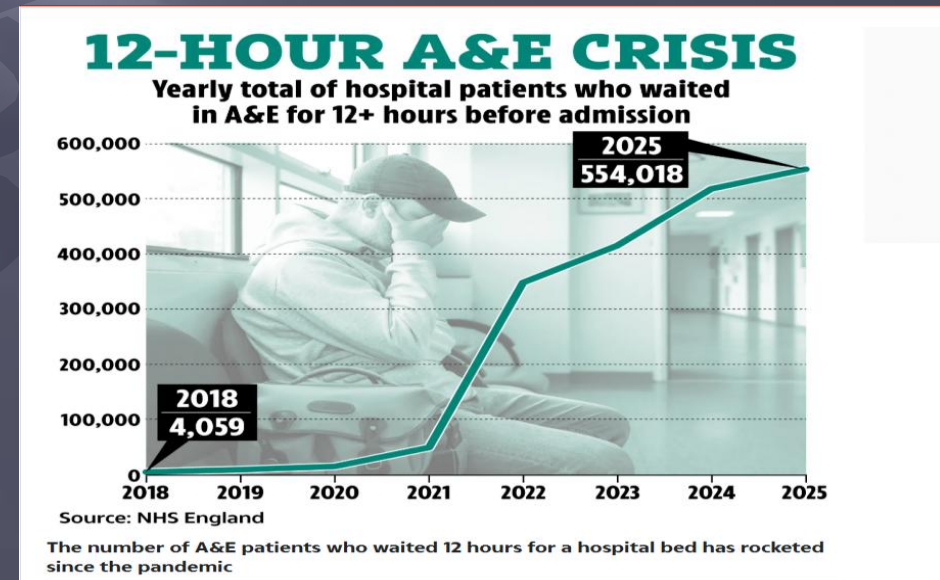
Health > News Health

'DEGRADING' Record half a MILLION patients endured 'torture' 12-hour A&E waits in 2025

'Trolley waits' are over 50 times more common than before Covid

Sam Blanchard, Health Correspondent
Published: 12:13, 15 Jan 2026

NHS patients lined up in A&E corridor



Research Gap

Existing System

- Hospitals use **reactive systems** that show current occupancy and delays. Limited use of machine learning for early prediction
- Most systems rely on **manual reports, static dashboards, or rule-based alerts.**
- Overcrowding is usually identified **after it has already happened.**
- Existing tools are often **hospital-specific** and not easily reusable.

What this project Add

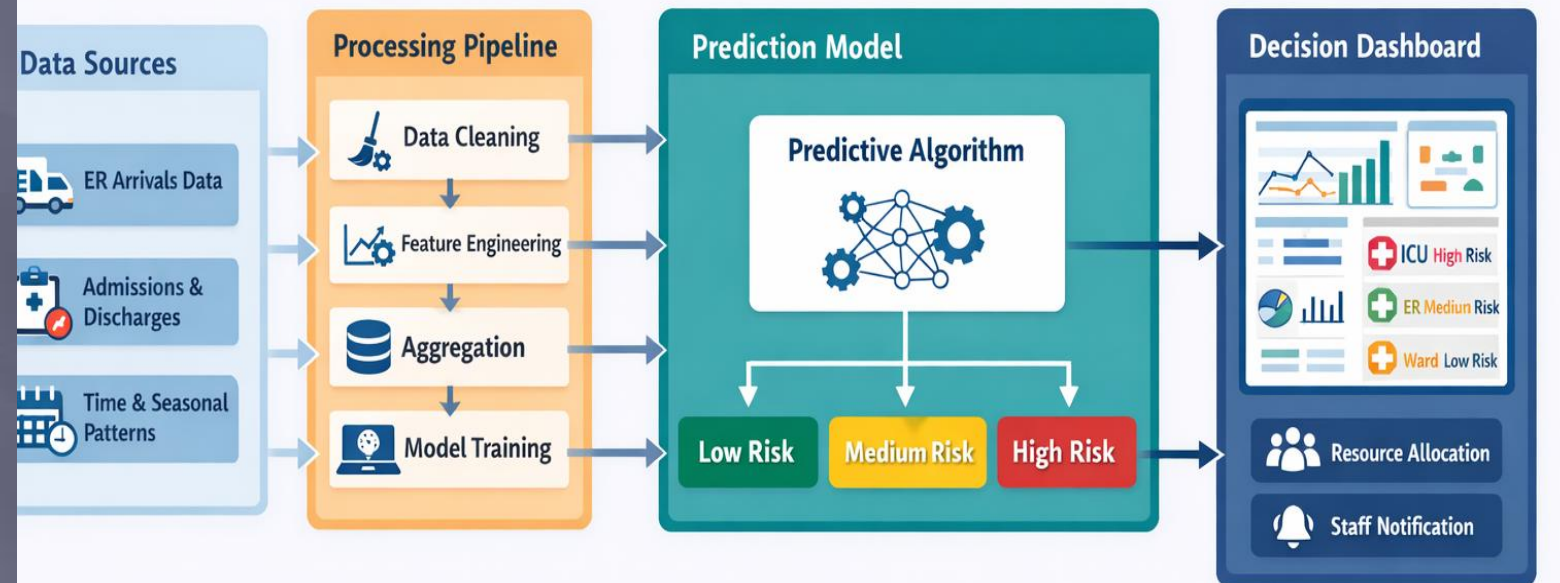
- A predictive approach instead of reactive monitoring
- **Early overload prediction** using historical patient flow patterns
- Focus on **department-level imbalance**, not just total hospital occupancy.
- Use of **leading indicators** (ER trends, discharge delays, stay duration).

Implementation Overview

I describe and implement this project based on **Load Risk**, not just raw numbers.

Load Risk is derived from:

- ER arrival trends
- Admission vs discharge gap
- Length of stay patterns
- Time factors (day, week, season)



Common Real-World Questions

How will my project address these questions

- How do we know if a department will be overcrowded tomorrow?
- Which department needs attention first?
- Why is Emergency overcrowded even when beds are available?
- Can we prepare for seasonal or recurring surges?
- How does this help decision-making, not just reporting?
- Can this system work for different hospitals?

This project does not replace doctors or hospital staff; it supports better operational decisions using data and AI.



Thank You